

Multiphonon Orbit-Lattice Relaxation of Excited States of Rare-Earth Ions in Crystals*†

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A systematic study has been made of multiphonon orbit-lattice relaxation of excited states of rare-earth ions in crystals. By measuring both excited-state fluorescence lifetimes and multiphonon relaxation quantum efficiencies, pure multiphonon transition rates were determined for the following set of crystals: (1) $\text{LaBr}_3\text{:Nd}^{3+}$, $\text{LaBr}_3\text{:Dy}^{3+}$, and $\text{LaBr}_3\text{:Ho}^{3+}$; (2) $\text{LaF}_3\text{:Er}^{3+}$ and $\text{LaF}_3\text{:Ho}^{3+}$; (3) $\text{SrF}_2\text{:Ho}^{3+}$ and $\text{SrF}_2\text{:Er}^{3+}$; and (4) $\text{Y}_2\text{O}_3\text{:Er}^{3+}$ and $\text{Y}_2\text{O}_3\text{:Ho}^{3+}$. Lifetimes were measured both in the near infrared using the fluorescence phase-shift technique, and in the visible region using pulse-excitation techniques. Measurements were performed at 4.2°K to determine spontaneous multiphonon transition rates, as well as at higher temperatures. The temperature dependences were fitted with a theoretical model involving the stimulated emission of phonons due to the thermally populated phonon modes. These results indicate that the decay involves the emission of high-energy optical phonons and occurs in approximately the lowest order permitted by energy conservation and cutoff in the phonon spectrum. The spontaneous transition rate was found to have a systematic exponential dependence on the transition energy to the terminal level, that is, on the energy gap. This behavior is interpreted as being due to the simple convergence of the orbit-lattice perturbation expansion. With the precise features of the interacting phonon modes and atomic levels averaged out for these high-order processes, the only critical parameter within a given lattice is the energy gap, i.e., the order of the process. The dependence on the order has been extracted by normalizing the energy gap by the optical-phonon cutoff energy. The rate for a given order and the swiftness of the convergence reflect the strength of the orbit-lattice interaction. The relative behavior for a given host can be explained through its known crystalline properties, in particular the phonon frequency distribution function and the static crystal field. The observed order dependence in Y_2O_3 indicates a relative weakness in the orbit-lattice interaction for crystals with strong covalency.

I. INTRODUCTION

THE problem of multiphonon orbit-lattice relaxation of excited states of rare-earth ions in crystals arose very early in the spectroscopic study of these materials. In observations of the fluorescence spectra, it was found that if two spin-orbit levels were sufficiently close in energy, the upper level would not be sufficiently stable to fluoresce. In LaCl_3 , for example, it was known that if a level was closer than about 1000 cm^{-1} to the level immediately below, it would not exhibit fluorescence.¹ The interpretation was that the energy was being emitted nonradiatively to the lattice as phonons. Since it was known that the cutoff in the phonon spectrum in LaCl_3 was of the order of a few hundred cm^{-1} , it was evident that the process was of a high order, involving several phonons. This critical energy gap was observed to be different in different crystal hosts.¹

In an early theoretical approach to the problem, Kiel^{2,3} carried the first-order term of the orbit-lattice interaction to high orders of perturbation theory to arrive at processes involving the simultaneous emission of many phonons. The problem was seen to be theo-

retically intractable, except for very rough order-of-magnitude estimates.

Later workers measuring excited-state lifetimes^{4,5} provided semiquantitative confirmation of what had been observed qualitatively. It was demonstrated that as the energy gap below a level became smaller, there was a tendency for the lifetime to decrease.

Pure multiphonon transition rates were determined by Partlow and Moos⁶ for $\text{LaCl}_3\text{:Nd}^{3+}$ and by Chamberlain, Paxman, and Page⁷ for the $^4S_{3/2}$ level of Er^{3+} in yttrium gallium garnet (YGG), by the measurement of both excited-state lifetimes and quantum efficiencies. Temperature-dependence measurements were fitted to a theoretical model involving the stimulated emission of phonons due to the thermally populated phonon modes, and indicated that the decay involved the emission of high-energy optical phonons.

Multiphonon transition rates have also been determined by Weber for several ions in LaF_3 ^{8,9} and Y_2O_3 ¹⁰ by the measurement of excited-state lifetimes and the subsequent subtraction of the calculated radiative transition rate from the measured value. The remainder is then interpreted to be the multiphonon decay contribution.

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¹ G. H. Dieke, in *Paramagnetic Resonance*, edited by W. Low (Academic Press Inc., New York, 1963), Vol. 2, p. 237.

² A. Kiel, Ph.D. dissertation, The Johns Hopkins University, 1962 (unpublished).

³ A. Kiel, in *Quantum Electronics*, edited by P. Grivet and N. Bloembergen (Columbia University Press, New York, 1964), Vol. 1, p. 765.

⁴ M. J. Weber, in *Physics of Quantum Electronics*, edited by P. L. Kelley, B. Lax, and P. E. Tannenwald (McGraw-Hill Book Co., New York, 1966), p. 350.

⁵ G. E. Barasch and G. H. Dieke, *J. Chem. Phys.* **43**, 988 (1965).

⁶ W. D. Partlow and H. W. Moos, *Phys. Rev.* **157**, 252 (1967).

⁷ J. R. Chamberlain, D. H. Paxman, and J. L. Page, *Proc. Phys. Soc. (London)* **89**, 143 (1966).

⁸ M. J. Weber, in *Optical Properties of Ions in Crystals*, edited by H. M. Crosswhite and H. W. Moos (Interscience Publishers, Inc., New York, 1966), p. 467.

⁹ M. J. Weber, *Phys. Rev.* **156**, 231 (1967); **157**, 262 (1967).

¹⁰ M. J. Weber, *Phys. Rev.* **171**, 283 (1968).

Multiphonon transition rates for the lower-lying levels of $\text{LaCl}_3:\text{Dy}^{3+}$ were measured by Riseberg, Gandrud, and Moos,¹¹ and it was found that the spontaneous multiphonon transition rate for a number of levels in LaCl_3 showed a systematic exponential dependence on the energy gap. Similar observations have been made by Weber for $\text{LaF}_3:\text{Er}^{3+}$ (Ref. 9) and $\text{Y}_2\text{O}_3:\text{Er}^{3+}$ (Ref. 10).

Along these lines, we have carried out a systematic study of the phenomenon of multiphonon orbit-lattice relaxation of excited states of rare-earth ions in crystals. Through the measurement of both excited-state lifetimes and multiphonon relaxation quantum efficiencies, multiphonon transition rates were determined for a number of excited states of rare-earth ions in several host crystals. Measurements at 4.2°K provided spontaneous transition rates, while measurements at higher temperatures could be analyzed using the stimulated-emission model, and gave information regarding the size of the phonons and the order of the process.

In addition to the data for LaCl_3 , some results have been reported for LaBr_3 and LaF_3 .¹² A number of temperature dependences indicate the general feature that the decay involves high-energy optical phonons in approximately the lowest-order process consistent with energy conservation and the cutoff in the phonon spectrum. The exponential dependence on the energy gap is systematic in each of the three crystals, and is interpreted as being due to the convergence of the appropriate perturbation expansion, with the precise features of the interacting phonon modes and ionic levels being statistically averaged out for processes involving many different phonon modes and equivalent processes. The relative behavior in each of the three cases can be explained through known properties of the host crystal, in particular, the static crystal fields and phonon spectra.

In this paper we report the results of this program. Detailed measurements and discussion for the rare-earth halides are presented, as well as for rare-earth ions in SrF_2 and Y_2O_3 . The data provide an invaluable tool for the prediction of multiphonon relaxation rates. Conclusions are drawn regarding the basic properties of multiphonon relaxation, as well as regarding general features of the orbit-lattice interaction.

After a brief recapitulation of the fundamental aspects of the theory in Sec. II, and a short discussion

of the experimental techniques in Sec. III, the experimental results and associated discussion are presented in Sec. IV. A general discussion of the results and the resulting conclusions are given in Sec. V.

II. THEORETICAL BACKGROUND

A. General

The interaction of an isolated rare-earth ion with its dynamical crystalline environment is interpreted as being due to the modulation of the crystalline electric field by the lattice vibrations. In a point-charge model, this is conceived of as the displacement of the point ions from their equilibrium positions. Expanding the crystal field in a Taylor's series about the equilibrium position, one can write for the orbit-lattice interaction

$$V_{o.l.} = V_e + \sum_i V_i Q_i + \frac{1}{2} \sum_{i,j} V_{ij} Q_i Q_j + \dots \quad (1)$$

Here, V_e is the static crystal field and the expansion is in terms of the normal-mode coordinates Q_i . The expansion coefficients V_i , V_{ij} , etc., are simply the partial derivatives of the crystalline field in the normal coordinate system;

$$V_i = \partial V_e / \partial Q_i, \quad V_{ij} = \partial^2 V_e / \partial Q_i \partial Q_j, \quad \dots \quad (2)$$

Thus a given term of the interaction Hamiltonian is a product of two terms, one of which operates on the atomic system and one on the lattice vibration system. Such an expansion is extremely difficult for all but the simplest crystals. The approach of Orbach¹³ for the calculation of spin-lattice relaxation rates involving acoustical phonons has been the most successful in avoiding this difficulty.

The occurrence of processes involving the emission of many phonons can be accounted for by considering the first-order term in the expansion and carrying this to high orders of perturbation theory. Kiel^{2,3} utilized this approach in his theoretical treatment of multiphonon transition rates. However, an additional source of multiphonon emission processes arises from high-order terms of $V_{o.l.}$ considered in first-order perturbation theory. Let us consider a transition between initial and final atomic states $|\psi_a\rangle$ and $|\psi_b\rangle$ with the emission of n phonons. The expression for the transition rate is due to a sum of these two sources, given by

$$\begin{aligned} W^{(n)} = & (2\pi/\hbar) \sum_{i \dots j, m_1 \dots m_{n-1}} |\langle n_i+1 | Q_i | n_i \rangle|^2 \dots |\langle n_j+1 | Q_j | n_j \rangle|^2 \\ & \times \frac{|\langle \psi_b | V_i | \psi_{m_{n-1}} \rangle|^2 \dots |\langle \psi_{m_1} | V_j | \psi_a \rangle|^2}{(E_{m_{n-1}} + \hbar\omega_j + \dots + \hbar\omega_{i-1} - E_a)^2 \dots (E_{m_1} + \hbar\omega_j - E_a)^2} \times g(\omega_i) \dots g(\omega_j) \delta(E_b + (\hbar\omega_i + \dots + \hbar\omega_j) - E_a) \\ & + (2\pi/\hbar) \sum_{i \dots j} (1/n!)^2 |\langle \psi_b | V_i \dots V_j | \psi_a \rangle|^2 |\langle n_i+1 | Q_i | n_i \rangle|^2 \dots |\langle n_j+1 | Q_j | n_j \rangle|^2 \\ & \times g(\omega_i) \dots g(\omega_j) \delta(E_b + (\hbar\omega_i + \dots + \hbar\omega_j) - E_a), \quad (3) \end{aligned}$$

¹¹ L. A. Riseberg, W. B. Gandrud, and H. W. Moos, Phys. Rev. **159**, 262 (1967).

¹² L. A. Riseberg and H. W. Moos, Phys. Rev. Letters **19**, 1423 (1967).

¹³ R. Orbach, Proc. Roy. Soc. (London) **A264**, 458 (1961).

where $n_i \dots n_j$ are phonon mode occupation numbers, $|\psi_{m_1}\rangle \dots |\psi_{m_{n-1}}\rangle$ are intermediate virtual states, and $g(\omega_i) \dots g(\omega_j)$ are the densities of phonon states. The atomic levels are assumed to be infinitely narrow, and energy conservation is guaranteed implicitly by the δ function. The most general form of this expression would also include the contributions from the intermediate terms. For example, an n th-order process can also arise from the terms of $V_{o.l.}$ of order $\frac{1}{2}n$ considered in second-order perturbation theory, as well as a number of other possibilities.

It is evident that *ab initio* calculations of such transition rates are quite intractable. One would have to consider all possible intermediate states, and all possible terminal levels (i.e., the Stark components of the multiplet to which relaxation is taking place). In addition, one must perform an n -fold convolution of the density of phonon states. Except for very rough order-of-magnitude estimates, there is little hope for success in the way of a theoretical solution, even if one had sufficient knowledge of the fundamental interaction.

However, because of the large number of interacting phonon modes and intermediate states, it is likely that any precise features of these modes and levels will be statistically averaged out for these high-order processes. This is supported by recent observations of the vibronic sidebands of electronic transitions in rare-earth chlorides.¹⁴ It was found that the main contribution arises from phonons corresponding to directions of low symmetry in the Brillouin zone. When one considers the orbit-lattice interaction at the rare-earth ion site (i.e., performs the reduction to the point group at the site of the rare earth), there is almost no symmetry even in this case for single-phonon processes. For multiphonon processes, it is therefore quite manifest that symmetry properties of the phonons will be relatively unimportant.

B. Temperature Dependence

Let us assume that a given multiphonon decay process involves the emission of p_i phonons of equal energy corresponding to the i th phonon mode. Then from Eq. (3) we can write

$$W(T) = W_0(n_i + 1)^{p_i}, \quad (4)$$

where n_i is the occupation number of the i th phonon mode and W_0 is the spontaneous transition rate [i.e., $W(T) = W_0$ at $T = 0$ when $n_i = 0$ and the phonon modes are all initially in their ground state]. As the temperature is raised the phonon modes become thermally populated and the multiphonon transition rate grows because of the stimulated emission of phonons. The assumption that the phonons involved are of equal energy is a reasonable approach, because even if the phonons had some spread about a central energy, the

behavior would not be very different. The critical feature in the temperature dependence is the order of the process.

The approximate energy gap to the next lowest level, ΔE , determines the energy of the phonons for a process of a given order as follows:

$$p_i \hbar \omega_i = \Delta E. \quad (5)$$

We now replace n_i by its Bose-Einstein average

$$\bar{n}_i = [\exp(\hbar \omega_i / kT) - 1]^{-1} \quad (6)$$

and finally obtain the temperature-dependent multiphonon transition rate

$$W(T) = W_0 \left(\frac{\exp(\hbar \omega_i / kT)}{\exp(\hbar \omega_i / kT) - 1} \right)^{p_i}. \quad (7)$$

This describes the decay from a rather idealized single-level model. Actually the decay in general emanates from a Stark multiplet with a range in energy of typically several hundred cm^{-1} . Since decay and excitation processes take place between these levels at rates of the order of 10^{10} – 10^{12} sec^{-1} (as is evident from linewidth studies¹⁵), the set of Stark levels comes to thermal equilibrium in times short compared with the multiphonon decay time. Since the decay rates are significantly slower for the upper levels due to the larger energy gaps, this depopulation of the bottom Stark level has a significant effect on the temperature dependence. What is observed as the total rate is the weighted average of the rates from the separate levels. That is, for a group of levels, the rate is given by

$$W = \frac{\sum_i W_i g_i \exp(-\Delta_i / kT)}{\sum_i g_i \exp(-\Delta_i / kT)}, \quad (8)$$

where W_i is the individual decay rate from the i th level, g_i is the degeneracy, and Δ_i is the energy separation from the bottom level.

With this model it is possible to obtain very good theoretical fits to the observed temperature dependences. Such considerations yield important information regarding the number and energy of the phonons involved in the decay, as will be seen in the discussion of the experimental results below.

C. Energy-Gap Dependence

The dependence of the multiphonon transition rate on the energy gap can be understood as a simple consequence of the convergence of the terms of the perturbation expansion. This can be demonstrated in a phenomenological way as follows: Consider first the ratio of the n th-order transition rate to the $(n-1)$ th-

¹⁴ E. Cohen and H. W. Moos, Phys. Rev. **161**, 258 (1967); **161**, 268 (1967).

¹⁵ W. M. Yen, W. C. Scott, and A. L. Schawlow, Phys. Rev. **136**, A271 (1964).

order transition rate in a given crystal,

$$W^{(n)}/W^{(n-1)} = \epsilon \ll 1, \quad (9)$$

where ϵ is a coupling constant which is necessarily small enough to ensure the reasonably swift convergence of the expansion. Kiel^{2,3} made a theoretical estimate of ϵ and concluded that it could be as small as 0.05. The rate for an n th-order process behaves approximately like

$$W^{(n)} \approx A\epsilon^n, \quad (10)$$

where A is some constant. If the decay is dominated by the lowest-order processes permitted (as is likely if there is fast convergence of the expansion), then the order is approximately determined by the energy gap according to the following expression:

$$n \approx \Delta E / \hbar \omega_{\max}, \quad (11)$$

where $\omega_{\max}$ is the frequency cutoff of the phonon spectrum. The most meaningful way of determining $\omega_{\max}$ is from observations of the vibronic sidebands accompanying sharp electronic transitions of rare-earth ions in the crystal. The vibronic spectra provide an effective phonon frequency distribution function for consideration of orbit-lattice interaction phenomena.¹⁶

Using (10) and (11), the dependence of the rate on the energy gap, disregarding any selection rules, is of the form

$$W = Ce^{\alpha \Delta E}, \quad (12)$$

where C and α are constants which are characteristic of the particular crystal. That this expression will validly relate the behavior of the transition rate for a number of different levels depends on to what extent the precise features of the individual phonons and the participating levels will be statistically averaged out due to the large number of phonons involved and the large number of equivalent processes. This exponential dependence of the transition rate on the energy gap is observed to describe quite generally the energy-gap dependence of multiphonon transition rates. This will be discussed below with specific reference to the experimental results.

III. EXPERIMENTAL TECHNIQUES

A. Crystals and Cooling

Single crystals of LaBr_3 : 1% Dy^{3+} , LaBr_3 : 1% Nd^{3+} , and LaBr_3 : 0.5% Ho^{3+} were grown in this laboratory under the supervision of E. F. Williams by a modified Bridgman technique first described by Hutchisson and Wong.¹⁷ These were encapsulated in Infrasil quartz tubes in an atmosphere of He at 600-mm pressure.

¹⁶ E. Cohen, L. A. Riseberg, and H. W. Moos, Phys. Rev. (to be published).

¹⁷ C. A. Hutchisson and E. Y. Wong, J. Chem. Phys. **29**, 754 (1958).

Single crystals of SrF_2 : 1% Er^{3+} , SrF_2 : 1% Ho^{3+} , LaF_3 : 1% Ho^{3+} , and LaF_3 : 0.5% Er^{3+} were obtained from Optovac, Inc. A single crystal of Y_2O_3 : 1% Er^{3+} was obtained from the Linde Company. In addition, a powder sample of Y_2O_3 : 1% Ho^{3+} was prepared from the oxalate by heating in oxygen.

Crystals were cooled to liquid-helium or liquid-nitrogen temperatures by direct immersion into the bath in a Dewar with a quartz tip. Cooling to temperatures between 77 and 295°K was provided by forcing dry nitrogen gas cooled by liquid nitrogen in a heat exchanger through a quartz flow tube containing the sample. Temperature was measured with a copper-Constantan thermocouple.

B. Infrared Fluorescence

Lifetimes of levels fluorescing in the near infrared were measured by the phase-shift technique. The infrared fluorescence apparatus has been described in detail elsewhere.¹¹ Selective excitation was provided by a premonochromator, and the pump beam was modulated by a mechanical chopper. The fluorescence was detected by a spectrometer fitted with a lead sulfide detector, and the phase shift measured with a lock-in amplifier.

C. Optical Fluorescence

For those levels fluorescing in the optical region, lifetime measurements were performed by means of pulse-excitation techniques. The source was a repetitively flashed E.G. and G. FX-31B flashlamp. Excitation wavelengths were selected by narrow-band interference filters. The fluorescence was observed with a 1-m Jarrell-Ash Czerny-Turner spectrometer fitted with either an EMI 9558 S-20 or RCA 7102 S-1 photomultiplier. The repetitive fluorescence waveform was averaged with a PAR TDH-9 waveform eductor.

D. Multiphonon Relaxation Quantum Efficiencies

Excitation spectra and relative absorption strengths were measured to determine the fraction of ions in a given excited level which decay to the next lowest level, that is, the multiphonon relaxation branching ratio or quantum efficiency. The technique for performing these measurements has been described by Partlow and Moos.⁶

The excitation spectra were measured using the optical and infrared fluorescence systems described above in steady-state operation. In the case of the optical fluorescence apparatus, a Bausch & Lomb 0.5-m $f/5$ spectrometer fitted with a tungsten filament lamp was used as a premonochromator in excitation.

Relative absorption strengths were monitored either with a Kodak Ektron lead-sulfide detector or with an Amperex 150 CVP S-1 photomultiplier. Both of these were calibrated in a relative sense against a C.M. Reeder thermopile.

TABLE I. Multiphonon orbit-lattice relaxation data in LaBr_3 at 4.2°K.

Ion	Level	Energy gap (cm^{-1})	Lifetime (μsec)	Multiphonon relaxation quantum efficiency	Transition rate (sec^{-1})
Dy	$B(^6H_{5/2})$	1030	3000 ± 300	0.20 ± 0.05	$(6.6 \pm 1.8) \times 10$
Dy	$C(^6F_{7/2})$	760	455 ± 200	~ 1.0	$(2.2 \pm 1.0) \times 10^3$
Dy	$E(^6F_{3/2})$	775	98 ± 8	0.90 ± 0.15	$[8.85(-2.6, +1.1)] \times 10^3$
Nd	$S(^2H_{9/2}, ^4F_{5/2})$	1000	155 ± 10	0.011 ± 0.001	$(7.45 \pm 1.0) \times 10$
Nd	$A(^4S_{3/2}, ^4F_{7/2})$	800	84 ± 6	0.39 ± 0.04	$(4.47 \pm 0.50) \times 10^3$
Ho	$I(^6F_1, ^6G_6)$	625	6.0 ± 2.0	~ 1.0	$(1.67 \pm 0.50) \times 10^3$

These measurements discriminate against all other energy transfer processes except those which directly involve relaxation between the two levels in question. Thus, all ion-pair and multiple-ion contributions can be eliminated by a judicious choice of the ions and levels to be studied. The radiative transition across the gap lies deep in the infrared and is generally not competitive with the multiphonon decay.

In a few experimental cases for small energy gaps, the fluorescence was very weak and the lifetime extremely fast. For these, it was reasonable to assume that the total decay rate was in fact the multiphonon decay rate, to a very good approximation, and hence it was not necessary to measure the quantum efficiency.

IV. EXPERIMENTAL RESULTS

A. LaCl_3

Multiphonon transition rates have been published for several levels of $\text{LaCl}_3:\text{Nd}^{3+}$ and $\text{LaCl}_3:\text{Dy}^{3+}$. These results will be used in the ensuing discussion and the

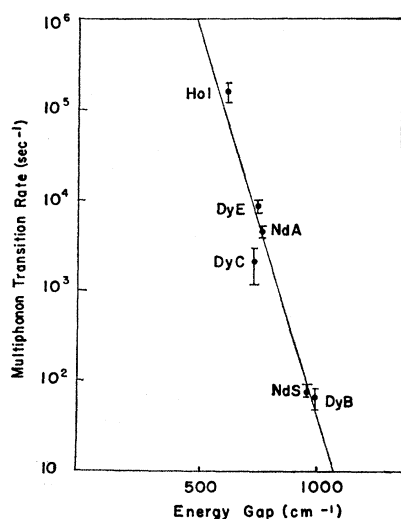


FIG. 1. Energy-gap dependence of the spontaneous multiphonon transition rate in LaBr_3 .

reader is referred to the literature^{6,11} for the details of these measurements.

B. LaBr_3

Multiphonon relaxation rates were determined for excited levels of $\text{LaBr}_3:\text{Nd}^{3+}$, $\text{LaBr}_3:\text{Dy}^{3+}$, and $\text{LaBr}_3:\text{Ho}^{3+}$. The structure is identical to that of the hexagonal rare-earth chlorides, and the rare-earth ion occupies a site of C_{3h} symmetry. Vibronic sidebands have been observed by Richman, Satten, and Wong¹⁸ for LaBr_3 , and indicate a cutoff in the phonon spectrum at approximately 175 cm^{-1} .

The temperature dependence of the multiphonon transition rate from $E(^6F_{3/2})$ to $D(^6F_{5/2})$ of Dy^{3+} in LaBr_3 was studied, and is shown in Fig. 1 of Ref. 12. We observe a flat portion at low temperatures due to the spontaneous emission of phonons and then the sharp break and increase at higher temperatures due to the stimulated emission to the thermally populated phonon modes. The theoretical fit corresponds to the emission of five 155-cm^{-1} phonons, given by

$$W(T) = 8.8 \times 10^3 \left(\frac{\exp[(155 \text{ cm}^{-1})hc/kT]}{\exp[(155 \text{ cm}^{-1})hc/kT] - 1} \right)^5 \text{ sec}^{-1}. \quad (13)$$

The unusual feature of this multiplet is that it consists of two levels which are separated by only a few cm^{-1} , and there is no effect due to the depopulation to upper levels with much slower transition rates. The fit to the pure five-phonon decay is therefore quite good. The decay involves the emission of phonons in the high-energy optical region.

The experimental results for those levels studied in LaBr_3 are given in Table I. The multiphonon transition rate is plotted as a function of energy gap in Fig. 1. We observe here a systematic exponential dependence of the transition rate on the gap, indicating the statistical averaging out of the precise features of the interacting phonon modes and atomic levels for high-order processes.

¹⁸ I. Richman, R. A. Satten, and E. Y. Wong, J. Chem. Phys. **39**, 1833 (1963).

TABLE II. Multiphonon orbit-lattice relaxation data in LaF₃ at 4.2°K.

Ion	Level	Energy gap (cm ⁻¹)	Lifetime (μsec)	Multiphonon relaxation quantum efficiency	Transition rate (sec ⁻¹)
Er	$L(^4G_{11/2})$	1550	6.2±0.5	0.75±0.20	(1.22±0.35)×10 ⁵
Er	$K(^2H_{9/2})$	1860	90±5	0.98±0.30	[1.09(-0.30, +0.05)]×10 ⁴
Er	$D(^4F_{9/2})$	2650	700±70	0.31±0.12	(4.4±1.5)×10 ²
Er	$B(^4I_{9/2})$	2025	135±20	~1.0	(7.4±1.1)×10 ³
Ho	$E(^5F_4, ^5S_2)$	2700	605±50	0.10±0.01	(1.69±0.20)×10 ²
Ho	$F(^5F_3)$	1800	32±2	0.72±0.07	(2.24±0.25)×10 ⁴
Ho	$K(^6G_4)$	1650	9.3±1.0	~1.0	(1.07±0.11)×10 ⁵
Ho	$J(^6G_6)$	1450	7±2	~1.0	(1.4±0.4)×10 ⁵

C. LaF₃

Multiphonon transition rates were determined for a number of levels of LaF₃:Ho³⁺ and LaF₃:Er³⁺. The structure here is different from the hexagonal chlorides and bromides. The symmetry at the rare-earth site is not known with certainty, but is thought to be C₂ or lower.¹⁹ Vibronic sidebands have been observed by Yen, Scott, and Schawlow¹⁵ and indicate that the energy cutoff of the phonon frequency distribution function is at approximately 350 cm⁻¹.

In Fig. 2 is shown the temperature dependence of the multiphonon transition rate from $K(^2H_{9/2})$ to $I(^4F_{3/2})$ of Er³⁺. The gap here is about 1860 cm⁻¹. The theoretical fit which has been drawn in is for the stimulated emission of six 310-cm⁻¹ phonons, with thermal depopulation to higher-lying Stark levels accounted for by the as-

sumption of three nondecaying levels 200 cm⁻¹ above the lowest. The expression for this dependence is given by

$$W(T) = 1.1 \times 10^4 \left(\frac{\exp[(310 \text{ cm}^{-1})hc/kT]}{\exp[(310 \text{ cm}^{-1})hc/kT] - 1} \right)^6 \{1 + 3 \exp[(-200 \text{ cm}^{-1})hc/kT]\}^{-1} \text{ sec}^{-1}. \quad (14)$$

It is evident that the fit to the data is extremely good, with only a few percent deviation for most points. The phonons involved are again observed to be in the high-energy optical region.

Figure 3 shows the temperature dependence of the multiphonon transition rate for the decay of $F(^5F_3)$ to $E(^5F_4, ^5S_2)$ of Ho³⁺ in LaF₃. The fit which has been drawn in represents the stimulated emission of six 300-cm⁻¹ phonons. Here, however, not only must the

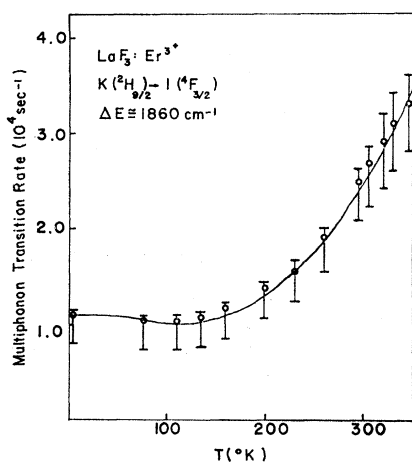


FIG. 2. Temperature dependence of the multiphonon transition rate from $K(^2H_{9/2})$ to $I(^4F_{3/2})$ of Er³⁺ in LaF₃. The theoretical curve indicates the expected temperature dependence for the emission of six 310-cm⁻¹ phonons with thermal depopulation to three nondecaying levels 200 cm⁻¹ above the lowest. The error bars indicate the absolute maximum probable error. The relative error between the points is much smaller.

¹⁹ W. F. Krupke and J. B. Gruber, J. Chem. Phys. **39**, 1024 (1963).

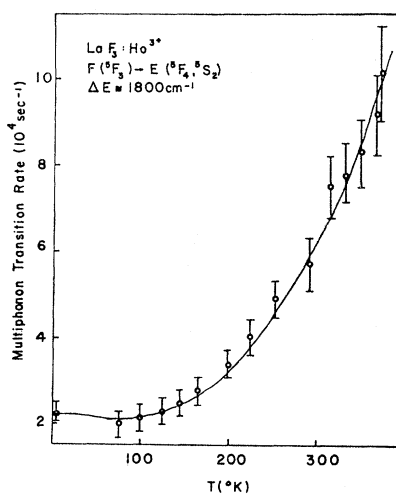


FIG. 3. Temperature dependence of the multiphonon transition rate from $F(^5F_3)$ to $E(^5F_4, ^5S_2)$ of Ho³⁺ in LaF₃. The theoretical curve indicates the expected temperature dependence for the emission of six 300-cm⁻¹ phonons with thermal depopulation to a single nondecaying level 150 cm⁻¹ above the lowest and five nondecaying levels 500 cm⁻¹ above the lowest. The error bars indicate the absolute maximum probable error. The relative error between the points is actually much smaller.

TABLE III. Multiphonon orbit-lattice relaxation data in SrF_2 at 4.2°K.

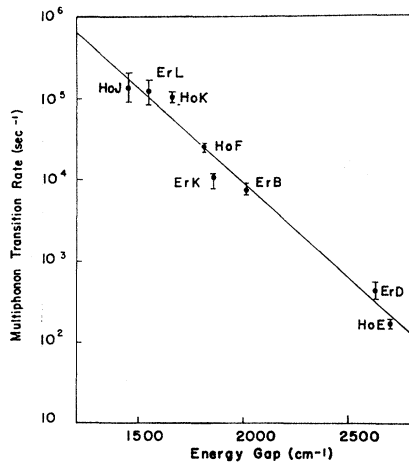
Ion	Level	Energy gap (cm^{-1})	Lifetime (μsec)	Multiphonon relaxation quantum efficiency	Transition rate (sec^{-1})
Er	$K(^2H_{9/2})$	1800	28.5 ± 2.0	0.77 ± 0.10	$(2.70 \pm 0.35) \times 10^4$
Er	$E(^4S_{3/2})$	3000	680 ± 50	0.16 ± 0.04	$(2.35 \pm 0.70) \times 10^3$
Er	$D(^4F_{9/2})$	2450	69 ± 7	0.51 ± 0.15	$(7.4 \pm 2.0) \times 10^3$
Er	$B(^4I_{9/2})$	2000	25 ± 5	~ 1.0	$(4.0 \pm 0.8) \times 10^4$
Ho	$E(^6F_4, ^6S_2)$	2600	160 ± 25	0.77 ± 0.20	$(4.81 \pm 1.68) \times 10^3$
Ho	$F(^6F_3)$	1750	7.3 ± 1.5	~ 1.0	$(1.37 \pm 0.25) \times 10^5$

depopulation to higher-lying Stark levels be considered but also depopulation to the next highest multiplets, $G(^6F_2)$ and $H(^8K_8)$, which lie only a few hundred cm^{-1} above F . Assuming a single nondecaying level at 150 cm^{-1} above the lowest to account for the higher Stark levels of F and five nondecaying levels at 500 cm^{-1} to account for the G and H multiplets, we arrive at the theoretical fit as shown, given by

$$W(T) = 2.2 \times 10^4 \left(\frac{\exp[(300 \text{ cm}^{-1})hc/kT]}{\exp[(300 \text{ cm}^{-1})hc/kT] - 1} \right)^6 \times \{1 + \exp[-(150 \text{ cm}^{-1})hc/kT] + 5 \exp[-(500 \text{ cm}^{-1})hc/kT]\}^{-1} \text{ sec}^{-1}. \quad (15)$$

Here, again, the theory provides an extremely good description of the experimental data. The decay involves phonons in the high-energy optical region, as was observed in the previous cases.

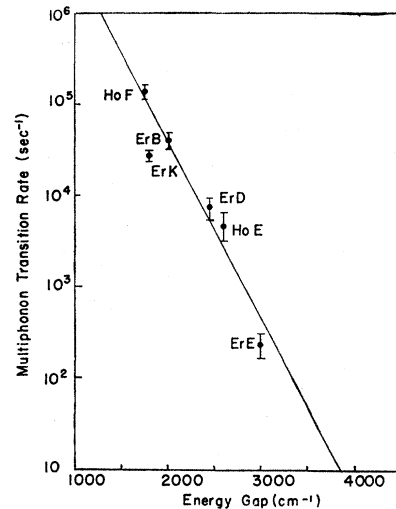
The multiphonon transition rates, with details of the measurements are given in Table II. Weber⁹ has also measured lifetimes in $\text{LaF}_3\text{:Er}^{3+}$ and our own results are in agreement with his. Our quantum efficiency measurements indicate that his radiative rate calculations are in many cases good to within $\pm 25\%$.

FIG. 4. Energy-gap dependence of the spontaneous multiphonon transition rate in LaF_3 .

A plot of the multiphonon transition rate as a function of energy gap in LaF_3 is shown in Fig. 4. Again we observe the systematic exponential dependence of the rate on the gap. The straight-line semilogarithmic behavior is extremely striking with deviations of less than 50% for most cases.

D. SrF_2

In SrF_2 , the trivalent rare-earth ion substitutes unnaturally for Sr^{2+} . Virtually all of the rare-earth dopant ions are found to be in either cubic or charge-compensated trigonal sites.²⁰ Since electric-dipole transitions are forbidden at the cubic site, the observed optical spectra can be unambiguously assigned to the trigonal site. Vibronic sidebands have been studied in SrF_2 by Richman²¹ and by Cohen and Guggenheim,²² and indicate a cutoff of the phonon frequency distribution function at approximately 360 cm^{-1} .

FIG. 5. Energy-gap dependence of the spontaneous multiphonon transition rate in SrF_2 .

²⁰ J. L. Merz and P. S. Pershan, Phys. Rev. **162**, 235 (1967).

²¹ I. Richman, Phys. Rev. **133**, A1364 (1964).

²² E. Cohen and H. J. Guggenheim, Bull. Am. Phys. Soc. **13**, 88 (1968).

TABLE IV. Multiphonon orbit-lattice relaxation data in Y_2O_3 at 42°K.

Ion	Level	Energy gap (cm^{-1})	Lifetime (μsec)	Multiphonon relaxation quantum efficiency	Transition rate (sec^{-1})
Er	$D(^4F_{9/2})$	2516	50 ± 15	0.84 ± 0.10	$(1.7 \pm 0.6) \times 10^4$
Er	$E(^4S_{3/2})$	2795	130 ± 6	0.82 ± 0.15	$(6.5 \pm 1.5) \times 10^3$
Er	$B(^4I_{9/2})$	1945	< 10	~ 1.0	$> 10^6$
Er	$K(^2H_{9/2})$	1871	5.4 ± 1.5	~ 1.0	$(1.85 \pm 0.5) \times 10^5$
Ho	$D(^6F_5)$	2100	15 ± 3	~ 1.0	$(6.7 \pm 1.4) \times 10^4$

Multiphonon transition rates were determined for a number of levels of $SrF_2:Ho^{3+}$ and $SrF_2:Er^{3+}$. The spontaneous transition rates measured, together with the details of the measurements, are given in Table III. A plot of the multiphonon transition rate as a function of energy gap is shown in Fig. 5. Again we observe the systematic exponential dependence of the rate on the energy gap, indicating the same type of averaging as in $LaCl_3$, $LaBr_3$, and LaF_3 .

E. Y_2O_3

The Y_2O_3 crystalline structure is isomorphous to that of the pure rare-earth oxides for the heavier ions, and thus Y_2O_3 accepts the rare earth as a natural substitution. The rare earth can enter at either of two inequivalent sites, with C_2 or C_{3i} point symmetry. Due to the inversion symmetry at the C_{3i} site, electric-dipole transitions are observed only for those ions in C_2 sites, and the optical spectrum can be interpreted in this way.²³

The vibronic sidebands associated with the electronic transitions are too weak to be observed readily in absorption. However, they are evident in excitation spectra at low resolution, and the cutoff in the phonon spectrum is observed to be at approximately 550 cm^{-1} .

Multiphonon transition rates were measured for several levels in $Y_2O_3:Er^{3+}$ and $Y_2O_3:Ho^{3+}$. The details of these measurements are given in Table IV. Weber¹⁰ has also performed measurements on $Y_2O_3:Er^{3+}$. Our lifetime measurements are substantially in agreement with his. The quantum efficiencies which we observe indicate excellent accuracy in his calculations of the radiative transition rates.

The spontaneous multiphonon transition rate in Y_2O_3 is plotted as a function of energy gap in Fig. 6. It is evident that the systematic exponential dependence is followed very closely.

V. DISCUSSION AND CONCLUSIONS

The temperature dependences that have been observed and fitted theoretically in $LaBr_3$ and LaF_3 exhibit the common feature that the decay involves

the lowest-order process consistent with energy conservation and the cutoff in the phonon spectrum. The temperature dependence of the multiphonon decay for several levels in $LaCl_3$ have also been analyzed using this model,^{6,11} with the result that the decay involves high-energy optical phonons and processes of approximately the lowest possible order.

Furthermore, it was found that subtle features of the temperature dependence such as depopulation could be readily explained on the basis of an extremely simplified model. That the depopulation effects in $Ho\ F$ and $Er\ K$ in LaF_3 have in fact been analyzed properly is strongly supported by the observed temperature dependence of $Dy\ E$ in $LaBr_3$. Here, in the latter case, where because of the small Stark splittings there can be no depopulation effects, the fit is very good for a simple stimulated emission model. It is likely, then, in view of the excellent agreement between theory and experiment in the other cases, that our analysis is correct. The observation that the lowest-order process dominates is indicative of the convergence of the multiphonon transition rate with order. This convergence with order is demonstrated in a striking manner by the

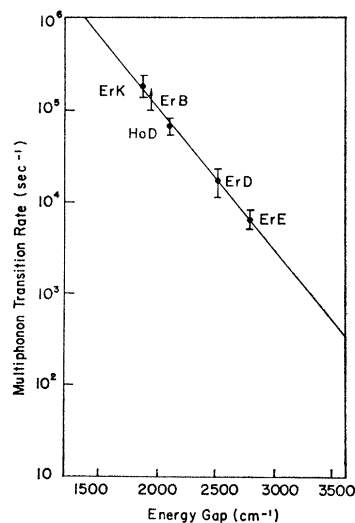


Fig. 6. Energy-gap dependence of the spontaneous multiphonon transition rate in Y_2O_3 .

²³ P. Kisluk, W. F. Krupke, and J. B. Gruber, J. Chem. Phys. **40**, 3606 (1964).

dependence of the transition rate on the energy gap that has been found for LaBr_3 , LaCl_3 , LaF_3 , SrF_2 , and Y_2O_3 . The systematic exponential dependence is a simple consequence of the convergence of the perturbation expansion as discussed in Sec. II. The statistical averaging out of the precise features of the interacting phonon modes and atomic levels is evidently a reasonable description. Since this is true, the use of (12) is valid for the determination of the multiphonon transition rate for any rare-earth level; i.e., within a given crystal host the critical parameter is the energy gap.

Thus, by means of the empirically determined energy-gap dependence, it is possible to predict multiphonon transition rates to a very high degree of accuracy. Examination of Figs. 1, 4, 5, and 6 indicates that such predictions can usually be made with better than 50% accuracy, and in all cases to within a factor of 3. A similar situation holds for LaCl_3 .^{10,11} This is quite important in view of the fact that *ab initio* calculations can be done with only order-of-magnitude accuracy, at best.

Of course, these statements must be qualified to exclude the class of $J=1$ to $J=0$ transitions, which have been discussed elsewhere as a special case due to the symmetry properties of the terms of the tensor-operator expansion of the orbit-lattice interaction.^{2,3,8-10,24}

The dependence on the order can be extracted from the energy-gap dependence by the approximate relationship of Eq. (11):

$$n \approx \Delta E / \hbar \omega_{\text{max}}$$

That is, if we normalize the energy gap by the energy cutoff in the phonon spectrum we obtain an "effective

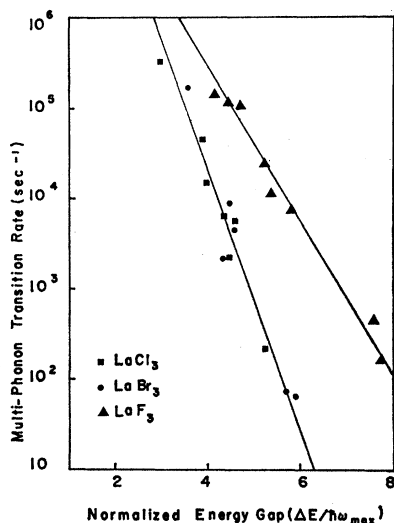


FIG. 7. Normalized energy-gap dependence of the spontaneous multiphonon transition rate for rare-earth halides. In LaCl_3 the energy gaps have been divided by 260 cm^{-1} , in LaBr_3 by 175 cm^{-1} , and in LaF_3 by 350 cm^{-1} .

²⁴ R. Orbach, *Phys. Rev.* **133**, A34 (1964).

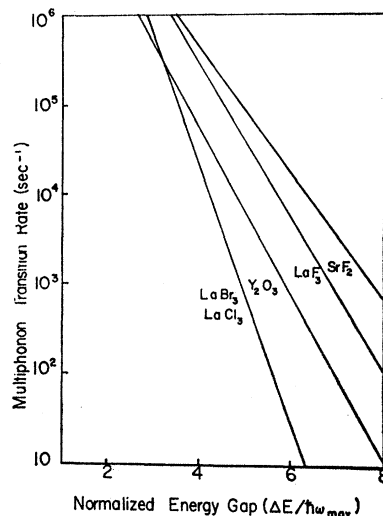


FIG. 8. Normalized energy-gap dependence of the spontaneous multiphonon transition rate for five crystals. In LaCl_3 the energy gaps have been divided by 260 cm^{-1} , in LaBr_3 by 175 cm^{-1} , in LaF_3 by 350 cm^{-1} , in SrF_2 by 360 cm^{-1} , and in Y_2O_3 by 550 cm^{-1} .

order" for the given process. Let us first consider the rare-earth halides, LaCl_3 , LaBr_3 , and LaF_3 . In LaCl_3 the gaps have been divided by the maximum phonon energy of 260 cm^{-1} , in LaBr_3 by 175 cm^{-1} , and in LaF_3 by 350 cm^{-1} . (The LaCl_3 data are from Refs. 6 and 11.) The multiphonon transition rates are shown plotted against this normalized energy gap in Fig. 7.¹²

The immediately striking feature is the exact collinearity of the LaBr_3 and LaCl_3 points. LaBr_3 and LaCl_3 have identical crystal structures, almost identical lattice parameters, and almost identical crystal fields.²⁵ The only difference with regard to the dynamic properties of the two crystals is the scaling down of the phonon spectrum due to the difference in the halogen masses. Thus, when we normalize the energy gap by the maximum phonon energy, we should observe exactly the same convergence with order for the two crystals. This is the situation of Fig. 7.

LaF_3 , on the other hand, has a different structure and a different phonon spectrum. The most significant difference for our consideration here is the larger static crystalline field of LaF_3 relative to LaBr_3 and LaCl_3 . The LaF_3 crystal-field splittings are typically twice as large as those in the chlorides and bromides. Under the reasonable assumption that the strength of the dynamic crystalline field reflects the strength of the static field, one would expect two main consequences: (1) The transition rate for a process of a given order would be faster in LaF_3 than in LaCl_3 and LaBr_3 , and (2) the convergence of the transition rate would be slower in LaF_3 than in LaCl_3 and LaBr_3 . These are precisely the features that are observed in Fig. 7.

²⁵ N. H. Kiess and G. H. Dieke, *J. Chem. Phys.* **45**, 2729 (1966).

TABLE V. Values of the coupling constant ϵ for five crystals.

Crystal	LaCl ₃	LaBr ₃	LaF ₃	SrF ₂	Y ₂ O ₃
ϵ	0.037 $\pm$ 0.004	0.037 $\pm$ 0.004	0.14 $\pm$ 0.01	0.20 $\pm$ 0.02	0.12 $\pm$ 0.02

The same considerations have been applied to the SrF₂ and Y₂O₃ data. The SrF₂ energy gaps have been normalized by 360 cm⁻¹ and the Y₂O₃ energy gaps by 550 cm⁻¹, the respective cutoffs in the phonon spectra. The results for the five crystals are shown plotted in Fig. 8.

SrF₂, which has slightly larger static crystalline-field splittings than LaF₃, is observed to exhibit the faster rates and slower convergence with order that would be expected on the basis of previous arguments.

Y₂O₃, on the other hand, has a much larger crystal field than any of the other hosts. The crystal-field splittings are typically 1½ to 2 times greater than in LaF₃, for example. Such large splittings are typical of crystals with strong covalency effects.

On the basis of the stronger crystalline field, one would expect faster multiphonon relaxation rates and a slower convergence of the rates with order than in any of the other crystals. But in Fig. 8, it is evident that the Y₂O₃ rates are in fact lower than would be expected, and the convergence faster. This behavior seems to indicate a relative weakness in the orbit-lattice interaction for strongly covalent crystals. This observation is consistent with the measurement of Chamberlain, Paxman, and Page⁷ on the ⁴S_{3/2} level of YGaG:Er³⁺. For this gap of approximately five phonons, they measured a transition rate of 5.3 $\times$ 10³ sec⁻¹. This rate, as is evident from Fig. 8, is much slower than would be expected on the basis of the strong static field in the garnets. This is also reasonable with regard to observations on the vibronic sidebands, which are relatively weaker for rare-earth ions in strongly covalent crystals.²⁶ This apparent weakness in the orbit-lattice interaction for strongly covalent crystals is a basically unsolved problem.

The coupling constant ϵ of Eq. (12), which determines the convergence properties for the particular crystal, can be extracted from Fig. 8. The values are given in Table V. Kiel's estimate^{2,3} that ϵ could be as small as 0.05 is certainly in agreement with what is observed experimentally.

²⁶ E. Cohen (private communication).

It must be pointed out at this juncture that this discussion must be qualified to apply to crystals such as CaWO₄. Here the problem is more subtle because of the presence both of lattice modes out to about 200 cm⁻¹ and internal vibrations of the tungstate complex, the most energetic of which is at 912 cm⁻¹.²⁷ We have investigated the decay of excited states of rare-earth ions in CaWO₄ and conclude that, as might be expected, the coupling to the internal vibrations of the tungstate group is weaker than the coupling to the lattice modes.²⁸ However, because of the presence of these different types of phonons with different energy ranges and coupling strengths, it is not possible to carry out a detailed analysis as in the cases discussed above.

In conclusion, this study has provided a great deal of information about the phenomenon of multiphonon orbit-lattice relaxation of rare-earth ions in crystals. The temperature dependence and the energy-gap dependence have indicated the convergence of the multiphonon transition rates with order. By knowing this behavior, it is possible to predict very accurately the multiphonon transition rate for a given level. The utility of this approach for other host crystals of interest is quite manifest. Furthermore, the dependence of the multiphonon transition rates on the order have been extracted from the energy-gap dependence, as in Fig. 8, yielding fundamental information regarding the orbit-lattice interaction, i.e., the transition rate for a given order and the swiftness of the convergence reflect the relative strength of $V_{o,l}$.

ACKNOWLEDGMENTS

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²⁷ S. P. Porto and J. F. Scott, Phys. Rev. **157**, 716 (1967).

²⁸ L. A. Riseberg, Ph.D. dissertation, The Johns Hopkins University, 1968 (unpublished).